

ab169558

Glycogen Assay Kit II (Colorimetric)

Instructions for Use

For the accurate measurement of glycogen levels in various biological samples.

This product is for research use only and is not intended for diagnostic use.

Table of Contents

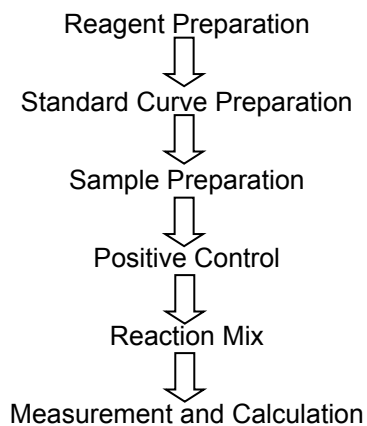
1.	Overview	3
2.	Protocol Summary	4
3.	Kit Components	5
4.	Storage	5
5.	Additional Materials Required	6
6.	Assay Protocol	6
7.	Data Analysis	9
8.	Troubleshooting	10

1. Overview

Glycogen serves as the main carbohydrate storage in animals and can be converted to glucose readily. It is primarily found in the liver and muscle tissues. Glycogen is a branched biopolymer comprising of α -1,4 linkage with α -1,6 linkages occurring every 8-10 glucose units along the backbone. Abnormal ability to utilize glycogen is found in diabetes and in several genetic glycogen storage diseases.

Abcam's Glycogen Assay Kit II (Colorimetric) (ab169558) provides a simple, fast and robust way to measure Glycogen levels in various biological samples. This assay is suitable for measuring Glycogen levels in samples that contain reducing substances, which may interfere with the oxidase-based assays. In this assay, Glycogen is hydrolyzed into glucose, which is oxidized to form an intermediate that reduces a colorless probe to a colored product with strong absorbance at 450 nm. This high-throughput suitable assay kit can detect less than 4 $\mu\text{g/ml}$ of Glycogen in samples.

2. Protocol Summary



3. Kit Components

Item	Quantity	Storage upon arrival	Storage after use/ reconstitution
Glycogen Hydrolysis Buffer	25 mL	-20°C	-20°C
Glycogen Development Buffer	25 mL	-20°C	-20°C
Hydrolysis Enzyme Mix (Lyophilized)	1 vial	-20°C	-20°C
Development Enzyme Mix (Lyophilized)	1 vial	-20°C	-20°C
Probe (Lyophilized)	1 vial	-20°C	-20°C
Glycogen Standard (2 mg/ml)	100 μ L	-20°C	-20°C

4. Storage

* Store the kit at -20°C and protect from light. Please read the entire protocol before performing the assay. Avoid repeated freeze/thaw cycles.

Warm assay buffers to room temperature before use. Briefly centrifuge all small vials prior to opening.

5. Additional Materials Required

- 96-well clear plate with flat bottoms
- Multi-well spectrophotometer (ELISA reader)

6. Assay Protocol

A. Reagent Preparation

1. Hydrolysis Enzyme Mix:

Reconstitute with 220 μ l Glycogen Hydrolysis Buffer. Pipette up and down to dissolve completely. Aliquot and store at -20°C . Avoid repeated freeze/thaw. Keep on ice while in use. Use within two months.

2. Development Enzyme Mix:

Reconstitute with 220 μ l ddH₂O. Pipette up and down to dissolve completely. Aliquot and store at -20°C . Keep on ice while in use. Use within two months.

3. Probe:

Reconstitute with 220 μ l dH₂O. Pipette up and down to dissolve completely. Stable for 2 months at -20°C .

B. Glycogen Assay Protocol

1. Glycogen Standard Curve:

Dilute Glycogen Standard to 0.2 mg/ml (0.2 µg/µl) by adding 10 µl of 2 mg/ml Glycogen Standard to 90 µl dH₂O, mix well. Add 0, 2, 4, 6, 8 and 10 µl of 0.2 mg/ml Glycogen Standard into series of wells in 96 well plate to generate 0, 0.4, 0.8, 1.2, 1.6 and 2 µg/well Glycogen Standard. Adjust volume to 50 µl per well with Glycogen Hydrolysis Buffer.

2. Sample preparation:

Tissue (10 mg) or cells (1×10^6) should be rapidly homogenized with 200 µl ddH₂O for 10 minutes on ice. Boil the homogenates for 10 minutes to inactivate enzymes. Centrifuge at 18000 rpm for 10 minutes and remove insoluble material. Collect the supernatant. Supernatant is ready to be assayed. Add 1-50 µl samples (~50 µg) into a 96 well plate and bring the volume to 50 µl with Glycogen Hydrolysis Buffer.

Notes: For unknown samples, we suggest testing several doses of your samples to ensure the readings are within the Standard Curve range.

Glucose in samples will generate background. If your sample has significant amount of glucose, a sample background control is required.

3. Hydrolysis:

Add 2 μ l of Hydrolysis Enzyme Mix to Standard and samples, mix well. Incubate at room temperature for 30 minutes.

Note: Don't add Hydrolysis Enzyme Mix to the sample background control.

4. Reaction Mix:

Mix enough reagents for the number of samples and standards to be performed. For each well, prepare a total 50 μ l Reaction Mix containing:

	Reaction Mix	Background Control Mix
Glycogen Development Buffer	44 μ l	46 μ l
Development Enzyme Mix	2 μ l	2 μ l
Probe	2 μ l	2 μ l

Add 48 μ l of the Reaction Mix to each well containing the Standard and samples and 50 μ l of Background Control Mix to background control well.

5. Measurement:

Incubate at room temperature for 30 minutes. Measure OD450 nm with a microplate reader.

7. Data Analysis

Calculation: Subtract 0 Glycogen Standard reading from all readings. Plot the Glycogen Standard curve. If background control reading is significant, subtract the background control reading from sample reading. Apply the corrected sample reading to the Glycogen Standard curve to get B μg of Glycogen in the samples.

$$\text{Sample glycogen concentration (C)} = \frac{B}{V} \times \text{Dilution Factor} = \mu\text{g}/\mu\text{l} = \text{mg/ml}$$

Where:

B is the Glycogen amount from Standard Curve (μg)

V is the sample volume used in the reaction well (μl).

Sample glycogen concentration can also be expressed in $\mu\text{g}/\text{mg}$ of sample or other desired method.

Glycogen molecular weight $\sim 10^5$ - 10^7 g/mol.

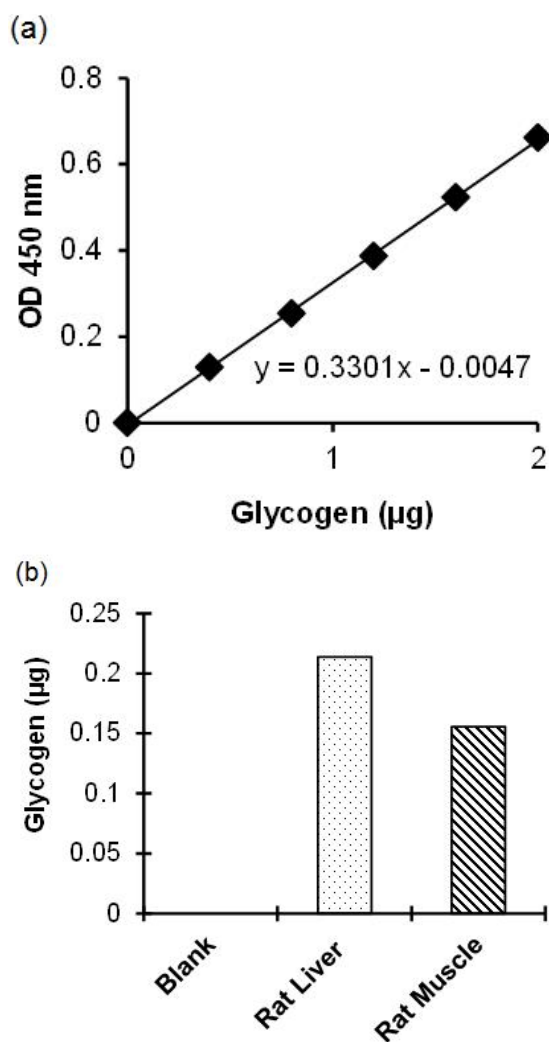


Figure 1: (a) Glycogen standard curve (b) Measurement of glycogen levels in rat liver (20 µg) and rat muscle (40 µg). Assays were performed following kit protocol.

8. Troubleshooting

Problem	Reason	Solution
Assay not working	Assay buffer at wrong temperature	Assay buffer must not be chilled - needs to be at RT
	Protocol step missed	Re-read and follow the protocol exactly
	Plate read at incorrect wavelength	Ensure you are using appropriate reader and filter settings (refer to datasheet)
	Unsuitable microtiter plate for assay	Fluorescence: Black plates (clear bottoms); Luminescence: White plates; Colorimetry: Clear plates. If critical, datasheet will indicate whether to use flat- or U-shaped wells
Unexpected results	Measured at wrong wavelength	Use appropriate reader and filter settings described in datasheet
	Samples contain impeding substances	Troubleshoot and also consider deproteinizing samples
	Unsuitable sample type	Use recommended samples types as listed on the datasheet
	Sample readings are outside linear range	Concentrate/ dilute samples to be in linear range
Samples with inconsistent readings	Unsuitable sample type	Refer to datasheet for details about incompatible samples
	Samples prepared in the wrong buffer	Use the assay buffer provided (or refer to datasheet for instructions)

Problem	Reason	Solution
	Cell/ tissue samples not sufficiently homogenized	Increase sonication time/ number of strokes with the Dounce homogenizer
	Too many freeze-thaw cycles	Aliquot samples to reduce the number of freeze-thaw cycles
	Samples contain impeding substances	Troubleshoot and also consider deproteinizing samples
	Samples are too old or incorrectly stored	Use freshly made samples and store at recommended temperature until use
Lower/ Higher readings in samples and standards	Not fully thawed kit components	Wait for components to thaw completely and gently mix prior use
	Out-of-date kit or incorrectly stored reagents	Always check expiry date and store kit components as recommended on the datasheet
	Reagents sitting for extended periods on ice	Try to prepare a fresh reaction mix prior to each use
	Incorrect incubation time/ temperature	Refer to datasheet for recommended incubation time and/ or temperature
	Incorrect amounts used	Check pipette is calibrated correctly (always use smallest volume pipette that can pipette entire volume)
Standard curve is not linear	Not fully thawed kit components	Wait for components to thaw completely and gently mix prior use
	Pipetting errors when setting up the standard curve	Try not to pipette too small volumes
	Incorrect pipetting when preparing the reaction mix	Always prepare a master mix

Problem	Reason	Solution
	Air bubbles in wells	Air bubbles will interfere with readings; try to avoid producing air bubbles and always remove bubbles prior to reading plates
	Concentration of standard stock incorrect	Recheck datasheet for recommended concentrations of standard stocks
	Errors in standard curve calculations	Refer to datasheet and re-check the calculations
	Use of other reagents than those provided with the kit	Use fresh components from the same kit

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